

Andrew Dover

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Incoming GenAI Software Engineering Intern, Micron Technology (ASIC Team) — Summer 2026

EDUCATION

San Jose State University

B.S. in Software Engineering

San Jose, CA

Dec. 2027

- **Organizations:** Software and Computer Engineering Society; Responsible Computing Club
- **Coursework:** Data Structures & Algorithms, Computer Organization & Architecture, Object-Oriented Design, Linear Algebra

EXPERIENCE

SJSU College of Engineering

Software Engineering Intern

San Jose, CA

Dec. 2025 – Jan. 2026

- Built monitoring service ingesting Prometheus metrics to detect missing time-series data and backfill gaps, restoring dashboard accuracy
- Improved CI/CD reliability with thread-safe deployment coordination using locks to prevent race conditions between concurrent deploys
- Modeled Prometheus API responses with Python dataclasses to enforce schema consistency and simplify metric processing

Software and Computer Engineering Society

Software Engineer, Development Team

San Jose, CA

Jan. 2026 – Present

- Developed BLETrack, a Bluetooth-based tracking system integrating into the club's infrastructure with custom network design and hardware integration
- Built and contributed features for internal club software used by 100+ members through pull requests, code reviews, and cross-team collaboration
- Diagnosed and resolved high-priority bugs across club systems, improving reliability for student users and developer experience

PROJECTS

Homelab Infrastructure Platform | *Proxmox, Docker, Authentik (SSO), Prometheus, Grafana, WireGuard, pfSense, Caddy*

- Designed segmented homelab using Proxmox, pfSense, and VLAN isolation to separate trusted home and lab services
- Built self-hosted platform with Docker, Caddy, and Authentik implementing centralized routing, TLS termination, and single sign-on
- Configured observability using Prometheus, Grafana, node-exporter, and cAdvisor, diagnosing container networking, DNS, and OAuth/OIDC issues

Liturgical.Display | *Python, FastAPI, Tailwind, Vosk, Redis, Docker Compose*

- Built scripture display system for local church with offline speech recognition, real-time web control, and synchronized WebSocket updates
- Architected multi-threaded runtime separating speech recognition from web serving to reduce latency and maintain responsiveness during live use
- Implemented concurrent audio pipeline with bounded queues for reliable real-time microphone input processing under continuous operation

Backcountry | *Python, FastAPI, React, Vite, LangGraph, Mapbox*

- Built full-stack backpacking trip planner with FastAPI backend, React frontend, and LangGraph-powered advisor for trip analysis and recommendations
- Engineered real-time data integration layer pulling from weather, air quality, fire perimeter, and snow-depth APIs with TTL caching for low latency
- Built deterministic risk engine synthesizing multi-source environmental data into actionable trip guidance with caution and no-go thresholds

Treat Your-shelf | *FastAPI, React, Firebase, LangChain, Actian Vector, Gemini*

- Developed AI pantry assistant converting pantry images and ingredient lists into recipe recommendations to reduce food waste
- Implemented persistent user profiles with Firebase Auth and Firestore for saved preferences and recipe filtering
- Integrated multimodal ingredient extraction with Gemini and semantic retrieval with Actian Vector for recipe suggestions

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, Lua, HTML/CSS

Frameworks: FastAPI, Flask, React, Next.js, Tailwind, LangChain, LangGraph

Infrastructure: Docker, Docker Compose, Linux, Proxmox, pfSense, WireGuard, Prometheus, Grafana, Caddy, Authentik

Tools/Databases: Claude Code, Claude Agent Management, Git, GitHub, PostgreSQL, Redis, Firebase, Mapbox, Actian Vector